

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Table of Contents

[PURPOSE](#)

[SCOPE](#)

[REFERENCES](#)

[TERMS, DEFINITIONS, ABBREVIATIONS AND ACRONYMS](#)

[DISCLAIMER](#)

[INTENT OF GUIDELINES](#)

[GUIDELINES ARE NOT](#)

[MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE](#)

Page 1 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

1. PURPOSE

This document describes the organization and structure of the Motor Control Transient Simulation Architecture for HA and VA Washers and how to configure for motor control algorithm simulation and tests. It is a live document and will be updated as more subsystems and variants or enhancement are added to the architecture.

2. SCOPE

The Motor Control Transient Simulation Architecture includes S- Function for Motor Control source code, Plant Models for BPM and CIM Motors, Rectifier Board Model, Inverter Model and interfaced together. In addition, architecture also includes parameter files and graphical user interface to set the architecture in particular configuration. The application can be in Vertical or Horizontal Axis washers.

3. REFERENCES

-

4. TERMS, DEFINITIONS, ABBREVIATIONS, AND ACRONYMS

DEFINITION	TERM
Controlled Induction Motor	CIM
Brushless Permanent Magnet	BPM
Motor Control Interface	MCI
Motor Control Unit	MCU
Vertical axis washer	VA
Horizontal axis washer	HA
High voltage direct current bus	HVDC
High voltage power supply	HVPS
Power factor correction	PFC
Functional Mockup Unit	FMU
Field Oriented Control	FOC

Page 2 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Direct Torque Control	DTC
Model In Loop	MIL
Software In Loop	SIL
Processor In Loop	PIL
Graphical User Interface	GUI

5. DISCLAIMER

The information in this document contains recommendations based on best practices available and have been proven to some extent in different products. Challenging these recommendations is possible, but some risk may be there and should be evaluated carefully.

5.1 Intent of Guidelines

Guidelines are intended to be living documents that improve over time based on a compilation of experiences. The document is concise and explains best practices, practices that should be avoided, and other points worth noting. Guidelines are an enabler of global cross-product knowledge sharing which can be useful for new, as well as highly experienced, employees

5.2 Guidelines Are Not

Guidelines are not a replacement for engineering rigor nor do they override consumer/regional requirements. Guidelines are not requirements and are therefore not specifications.

6. Motor Control Transient Simulation Architecture

Motor Control Transient Simulation architecture (Architecture onwards) is a toolset and frame-work for transient simulation of motor control loop modeled in Simulink in order to simulate the motor control algorithms in Model in Loop (MIL), Software in Loop (SIL).

6.1 System Requirements:

1. Matlab R2016b 64-bit
2. Compiler- Microsoft Visual C++ 2015 Professional (C)

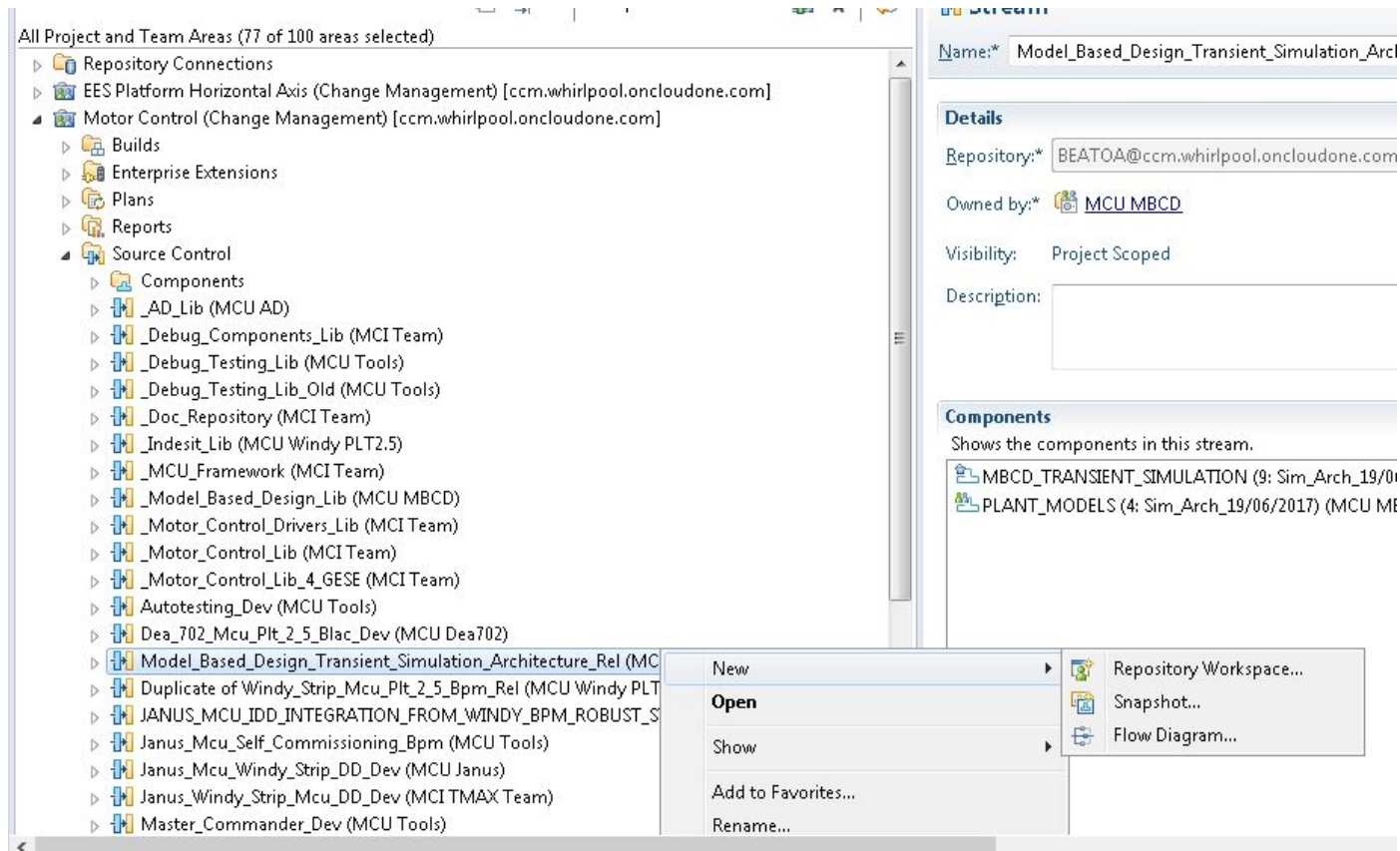
Page 3 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

3. FMKit (This toolbox will be invoked implicitly when the model is loaded)

6.2 Loading Architecture from RTC

1. Go to Motor Control (Change Management) Area and right click on the stream Model_Based_Design_Transient_Simulation_Architecture_Rel
2. Select New Repository Workspace



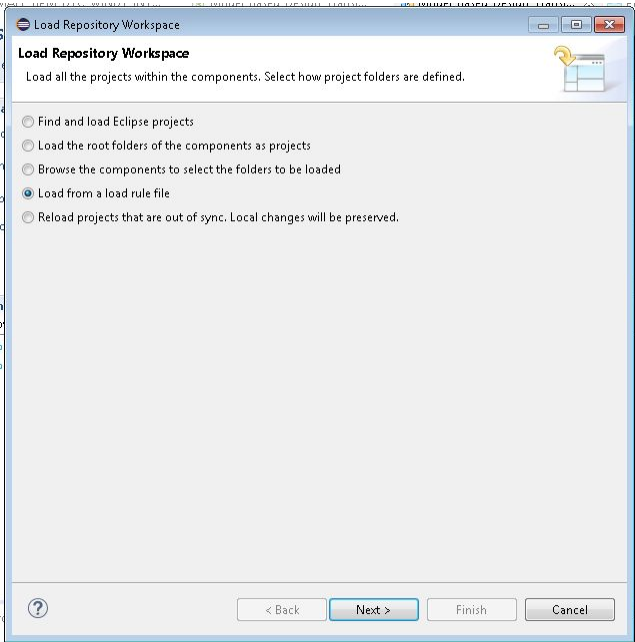
The screenshot shows the EES software interface. On the left, the 'All Project and Team Areas (77 of 100 areas selected)' tree is expanded to 'Motor Control (Change Management) [ccm.whirlpool.oncloudone.com]'. The 'Source Control' folder is expanded, showing a list of components. The component 'Model_Based_Design_Transient_Simulation_Architecture_Rel (MCU M...' is selected. A right-click context menu is open over this component, with the 'New' option selected, which has opened a sub-menu. In this sub-menu, the 'Repository Workspace...' option is highlighted. On the right side of the interface, the 'Details' panel for the selected component is visible, showing the repository name 'BEATO@ccm.whirlpool.oncloudone.com', the owner 'MCU MBCD', and the visibility 'Project Scoped'. The 'Components' panel below it shows a list of components in the stream, including 'MBCD_TRANSIENT_SIMULATION (9: Sim_Arch_19/0)' and 'PLANT_MODELS (4: Sim_Arch_19/06/2017) (MCU M...'.

3. Choose a name for the workspace and click on Finish

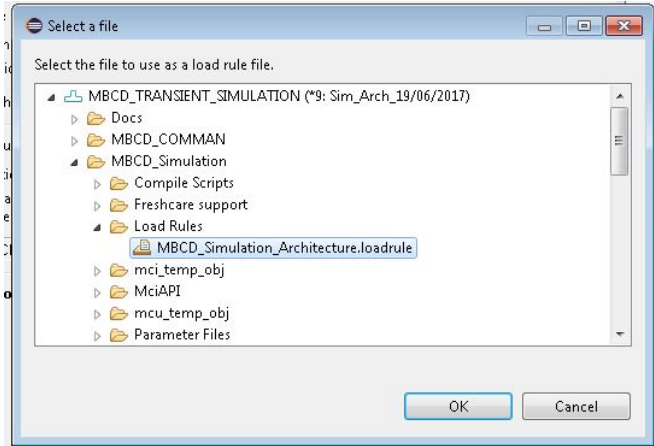
4. Select load from a load rule file and click Next

Page 4 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



5. Select Load from a remote load rule file. The load rule is located here. Click Ok.



Page 5 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

6. The Simulation Architecture will be loaded into your sandbox.

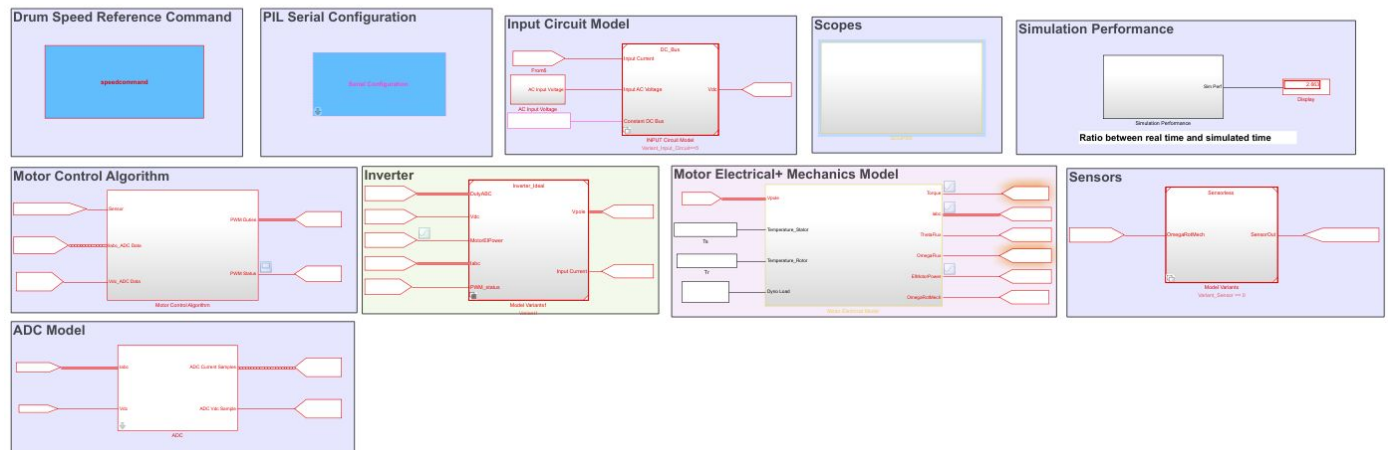
6.3 Organization of the Architecture

Architecture is structured into three layers.

1. Layer-1: System Layer

This is the topmost layer where all the subsystems of the motor control system are interfaced as shown below:

Motor Control Transient Simulation Architecture



2. Layer-2: Sub-System Layer

Sub-System layer consists of Model Variants for each sub-system of the motor control system. Simulink Model Variant is used to model variants of each sub-system in a single system model.

Following table describes the variants for each- subsystem

Subsystem	Variants
Input Circuit	Constant DC Bus
	HVPS 120V
	HVPS 120PFC
	HVPS 230
	HVPS 230 PFC
Motor+Mechanics Coupled	CIM Basic
	CIM MBCD
	BPM Basic-HA

Page 6 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

	BPM Basic-VA
	BPM MBCD- VA Agitation
	BPM MBCD- VA Spin
Sensor	Tachometer
	Sensorless
Inverter	Ideal Inverter
ADC Model	-
Control Model	SIL
	PIL
Signal Inspector	-
Simulation Performance	-

Variant Control Parameters for each sub-system and variant selection conditions are as listed below:

Variant Control Parameter	Variant Control Condition	Sub-System Variant Invoked in the System Model
Variant_Input_Circuit	1	HVPS 120 V
	2	HVPS 120 PFC
	3	HVPS 230
	4	HVPS 230PFC
	5	Constant DC Bus
Variant_Motor_Control	1	SIL
	2	PIL
Variant_Motor	1	HA-CIM-Basic Model
	2	HA-CIM-MBCD (Advanced Model)
	3	HA-BPM-Basic Model
	4	HA-BPM-MBCD(Advanced Model)
	5	VA- BPM-Agitation Model
	6	VA-BPM-Spin Model
	7	VA-BPM-Basic Model
Variant_Sensor	0	Sensorless
	1	Tachometer Model

3. Layer-3: Component Layer

Component layer consists of the models of the components either developed in Simulink or developed in Dymola and imported in Simulink as FMUs. Sub-System Plant Models are separate Simulink models

Page 7 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

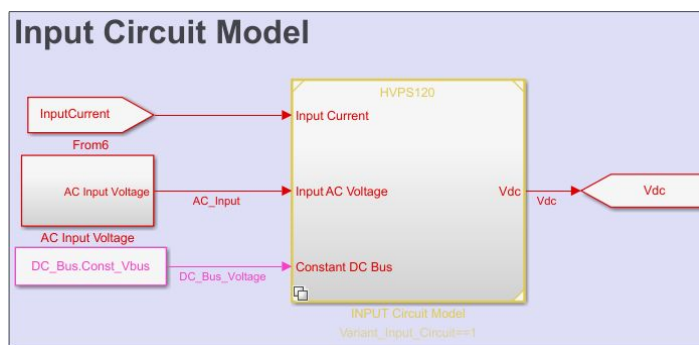
Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

referenced into the system model using Model Variant blocks.

Component	Model Source	Import Method In Simulink
Input Circuit Model	Dymola Model	FMU Co-Simulation using Ccode Solver. Imported using FMI toolkit.
CIM Basic Model	Simulink Model	-
CIM Advanced Model	Dymola Model	FMU Co-Simulation using Ccode Solver. Imported using FMI toolkit.
BPM Basic Model	Simulink Model	-
BPM Advanced Models	Dymola Model	FMU Co-Simulation using Ccode Solver Imported using FMI toolkit.
Sensor	Simulink Model	-
Control Model	Embedded Source Code	S- Function to interface production code into Simulink.

Short Description about the sub-systems and components of the Motor Control transient simulation architecture is described below:

3.1 Input Circuit Model:

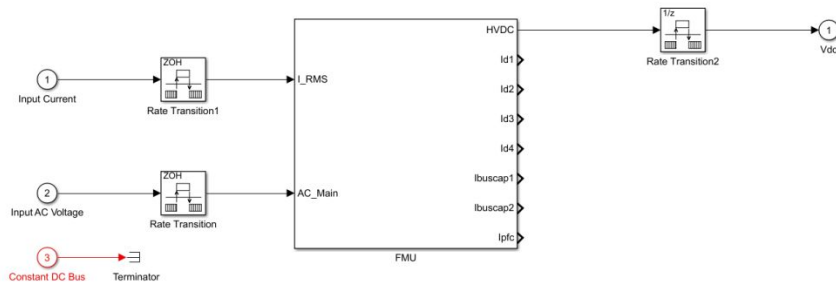


This Model converts AC Input Voltage to the Constant DC Bus Value which in turn acts as the input

Page 8 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

to the Control Model and 3-Phase Inverter model. Rectifier Variants for the input circuit are modeled as FMUs for each particular board families as follows:



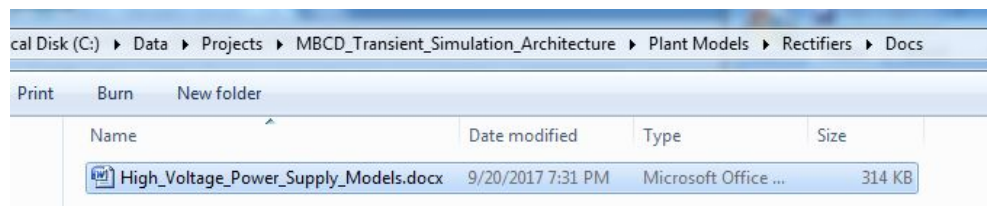
Inputs:

1. Inverter Input Current (A)
2. AC Input Voltage (V)
3. Constant DC Bus Voltage(For the Simulation of Constant DC Bus Simulation) (V)

Outputs:

1. DC Bus Voltage (V)

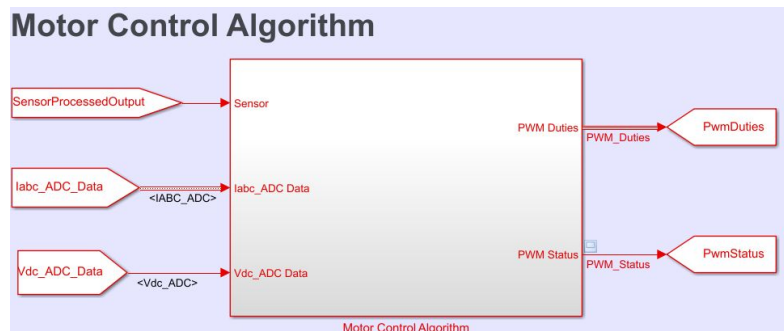
Detailed documentation for the Rectifier Models is available in the Plant Model folder as shown below:



3.2 Motor Control Algorithm

Page 9 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



This Model implements the Motor Control Algorithm. In SIL Simulation, Motor Control Source Code is compiled to the S-Function. In PIL Simulation, this model acts as interface in between evaluation/production board and Simulink environment.

Inputs:

1. Sensor Output (Output of the Speed Sensor)
2. ADC Data for Motor 3-Phase Currents
3. ADC Data for the DC Bus Voltage.

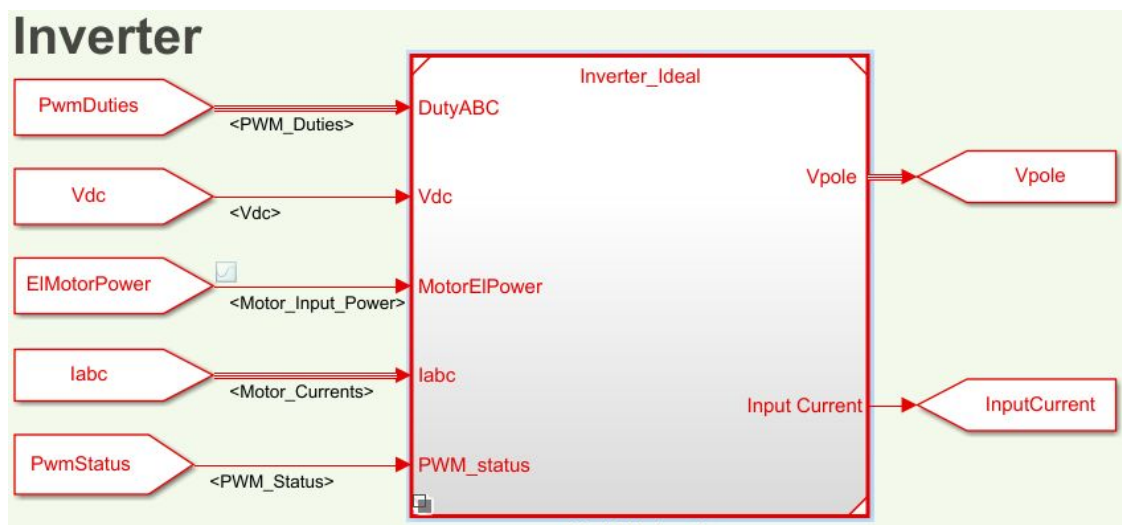
Outputs:

1. PWM Duty Cycles for 3-Phase Inverter. (%)
2. PWM Status(1/0)

3.3 Inverter

Page 10 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



Inverter Model implements the ideal 3-Phase inverter model. Inverter model generates the 3 pole voltages corresponding to each leg of the 3-phase inverter. These voltages are applied to the motor models.

Inputs:

1. PWM Duty Cycles for 3-Phase Inverter. (%)
2. DC- Bus Voltage. (V)
3. Motor Input Power(W)
4. Motor 3-Phase Input Currents(A)
5. PWM Status(0/1)

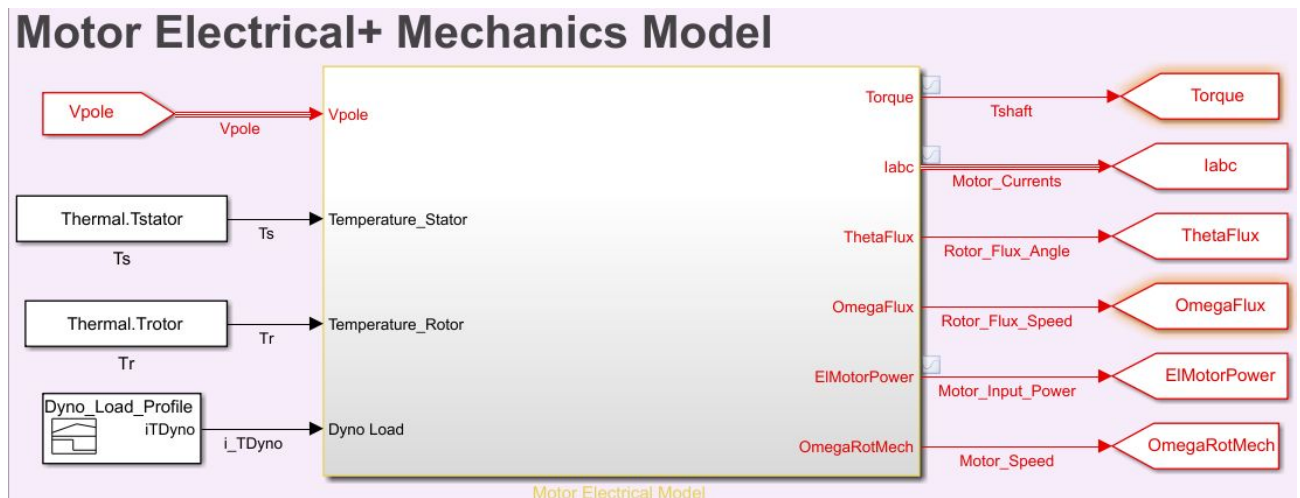
Outputs:

1. Pole Voltages for 3 Inverter Legs.
2. Input Current from the DC- Bus

3.4 Motor Electromechanical Model

Page 11 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



Motor Electro Mechanical Model implements the Models for the Electric Motor and the mechanical components which are part of the washer drive system. By keeping the interfaces same, various motor and drive system/load models are invoked as per the architecture configuration.

Inputs:

1. Pole Voltages (V). (Model Internally converts into the Motor Phase Voltage)
2. Temperature of the Stator Windings (Deg. C)
3. Temperature of the Rotor Windings/Magnets (Deg. C)
4. Dyno Load Torque. (N.m)

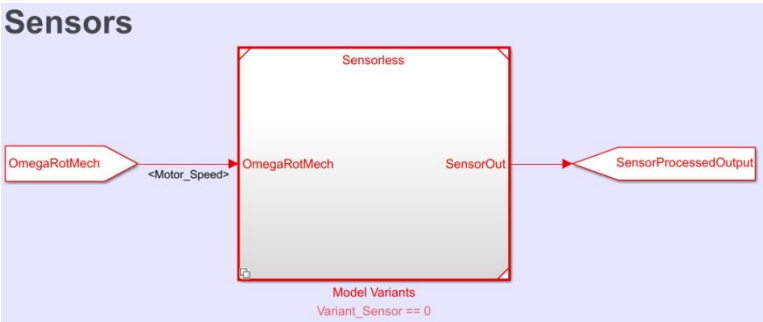
Outputs:

1. Motor Shaft Torque (N.m)
2. 3-Phase Motor Currents (A)
3. Rotor Flux Position (rad)
4. Rotor Flux angular speed. (rad/sec)
5. Motor Input Power(W)
6. Motor angular speed. (rad/sec)

Advanced model variants are modeled as FMUs from Dymola as follows:

Page 12 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



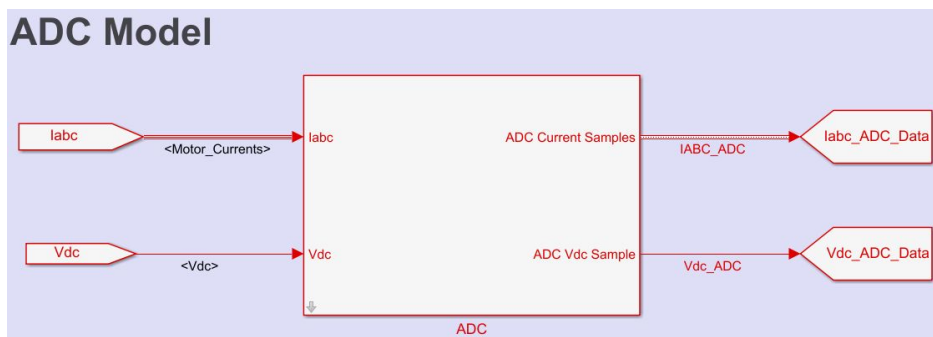
Sensor model consists of the model of the tachometer in case of CIM projects and where as this Sensorless model is invoked in case of all BPM Project In case of Sensor Model invoked; Sensor Output will consist of signal as per the following Tachometer model:



3.6 ADC Model:

Page 14 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



ADC Model converts Physical Feedback signals into corresponding ADC Levels. This ADC Signals are in turn input to the control model. Appropriate parameters will be invoked when Input Board is selected.

Inputs:

1. 3-Phase Motor Currents. (A)
2. DC Bus Voltage (V)

Outputs:

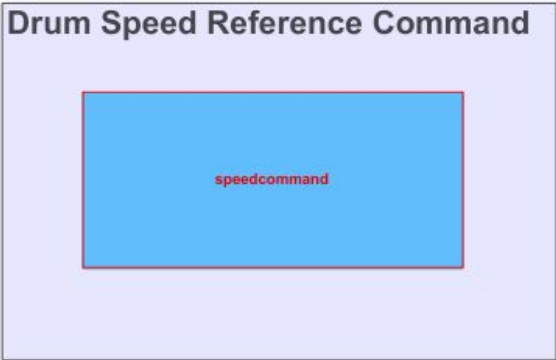
1. ADC Data corresponding to the 3-Phase Motor Currents.

4. Speed Command UI:

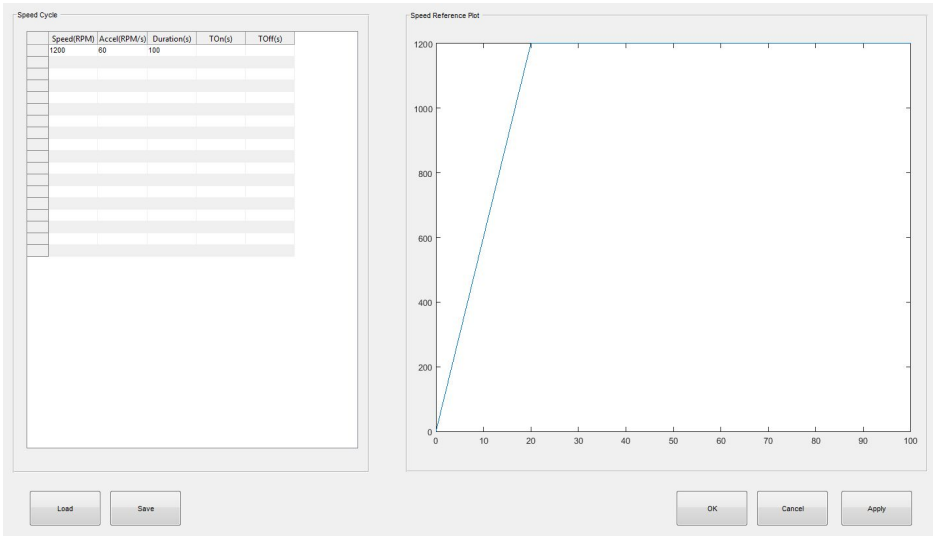
Different speed commands for the Drum can be generated including wash and spin cycles with the help of Speed Command UI as shown in figure below:

Page 15 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



Speed Command UI can be accessed by double clicking on the speed command S-Function block:



Speed Commands can be saved as Saved as .MAT file using the Save Pushbutton on the speed command UI. Previously saved speed command can be loaded by using the Load pushbutton on the speed command UI.

5. Model Configuration UI:

Architecture can be configured for different combinations of the sub-systems with the help of **Sim Arch Config UI** as shown below:

Page 16 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Motor Control Transient Simulation Architecture

Control Model Variants

☒ SIL-SRMotor-MCI

☐ PL-SRMotor-MCI

Motor Control Options

☒ Fast ADC Initialization

☒ Skip Motor Moving Handler

Select Control Source

☒ Source

Select Source Folder

☐ Pre-Compiled Objects

Select Object Folder

MCI Sampling Frequency

kHz

Select Board

Windy Full h60_WithPFC.xml
Windy International_230VWithPFC.xml
Windy Strip h40_WithPFC.xml
Windy Strip h50_WithPFC.xml

Input Circuit Board Variants

☒ Constant DC Bus Voltage

325 V

☐ Rectifier Model Variant

Rectifier Variant

☐ 120V

☒ 230V

PFC Variant

☐ With PFC

☒ Without PFC

Rectifier Model Frequency

☐ 2kHz

☐ 0.5/MCI Frequency

☒ MCI Frequency

Motor Model Variants

Select Motor

☐ CM

☒ BPM

Model Fidelity

☒ Basic Model

☐ Advanced Model

BPM Variant

☒ sBPM

☐ gBPM

VA Mode

☒ Agitation

☐ Spin

Motor Power Loss

☐ Power Loss Enabled

Motor Temperature

Tstator 22 DegC

Trotor 22 DegC

Nidec_38mm.xml
Nidec_38mm.xml
Nidec_42mm.xml
Nidec_53mm.xml

Mechanical Parameters

Mechanical HA.xml

Mechanical Model Variant

☐ Dyno Load

☒ Washer Drive System Load

Drum Properties

☐ Enable

☒ Disable

Weight of Drum(M_D) Kg

Weight of Inertial Mass(M) Kg

Radius of Drum(R_Drum) m

Thickness of Inertial Mass (R_J) m

Calculate Drum Inertia

Simulate Base Model Save Base Workspace

Different Model configuration options using this UI have been explained in the section 6.4

6. Parameter Files

Each Sub-System in the architecture is parameterized. Parameters for the plants are maintained in the XML documents in the **Parameter Files** Folder:

cal Disk (C:) ▶ Data ▶ Projects ▶ MBCD_Transient_Simulation_Architecture ▶ Parameter Files ▶			
Share with ▼ Burn New folder			
Name	Date modified	Type	S
 Mechanical Parameters	9/20/2017 7:31 PM	File folder	
 Motor Parameters	9/20/2017 7:31 PM	File folder	
 Rectifier Parameters	9/20/2017 7:30 PM	File folder	

Page 17 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Parameter XML File example:

```

Cyborg_120.xml
1  <?xml version="1.0" encoding="UTF-8"?>
2  <Board_Parameters>
3      <R_line>0.8</R_line>
4      <L_line>0.00005</L_line>
5      <C_buscap>0.00100</C_buscap>
6      <R_diode>0.08</R_diode>
7      <R_buscap>0.16579</R_buscap>
8      <V_knee>0.70</V_knee>
9      <G_off>1.67E-08</G_off>
10     <L_pfc>0</L_pfc>
11     <R_pfc>0</R_pfc>
12     <Vr>120</Vr>
13     <Vdc_i>300</Vdc_i>
14     <freq>60</freq>
15     <C1>-0.186419</C1>
16     <C2>1.69504</C2>
17     <Cvlt>0.00636329</Cvlt>
18     <Vcc>3.3</Vcc>
19     <ADCBit>12</ADCBit>
20     <BoardVariant>Cyborg_120</BoardVariant>
21 </Board_Parameters>

```

Parameters are loaded as Structures (except variant control) corresponding to each sub-system in the base workspace as shown below:

Page 18 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

 config	1x1 struct
 DC_Bus	1x1 struct
 InputCircuit	1x1 struct
 Mechanical	1x1 struct
 Motor	1x1 struct
 Paths	1x1 struct
 Thermal	1x1 struct
 TimeSteps	1x1 struct
 Variant_Input_Circuit	5
 Variant_Motor	3
 Variant_Motor_Control	1
 Variant_Sensor	0

'config' structure is populated with the configuration and variant details for all the sub-systems as shown:

 project	'New Project'
 platform	'HA'
 Motor	'BPM'
 platform_flag	'HA'
 MCL_Sampling_Frequency	'Set'
 rect_flag	0
 const_dcbus_flag	1
 Root_Path_flag	'Source Selected'
 Motor_Control_Source	'Set'
 Control_Layer	'SIL'
 Fast_ADC_Initialization_Flag	'Set'
 Skip_Motor_Moving_Handler_Flag	'Set'
 Mech_Flag	0
 MotorVariant	'DDBPM_Janus_5.0cuft'
 MechanicalVariant	'HA_Spin'
 BoardVariant	'Windy International_230WithPFC'

Page 19 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

'Motor' structure in the base workspace populates the parameters related to the Motor Plant Model:

Motor	
1x1 struct with 4 fields	
Field	Value
Electrical	1x1 struct
Mechanical	1x1 struct
Power_Loss	1x1 struct
Inductance_Variation	1x1 struct

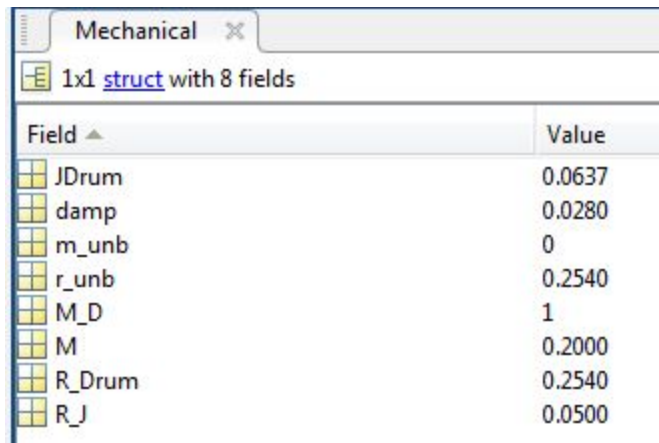
'InputCircuit' structure in the base workspace populates the parameters related to the Input Circuit Model and the ADC Model:

InputCircuit	
1x1 struct with 17 fields	
Field	Value
R_line	0.8000
L_line	5.0000e-05
C_buscap	5.6000e-04
R_diode	0.0790
R_buscap	0.3550
V_knee	0.7000
G_off	8.3300e-09
L_pfc	0.0071
R_pfc	0.5500
Vr	230
Vdc_i	300
freq	50
C1	-0.1553
C2	1.6886
Cvolt	0.0064
Vcc	3.3000
ADCBit	12

Page 20 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

'Mechanical' structure in the base workspace populates the parameters related to the Motor Plant Model:



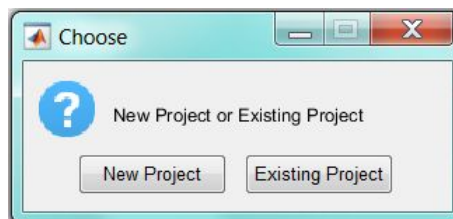
Field	Value
JDrum	0.0637
damp	0.0280
m_unb	0
r_unb	0.2540
M_D	1
M	0.2000
R_Drum	0.2540
R_J	0.0500

6.4 Configuration of the Architecture for the Simulation

Following steps will describe the procedure to work with the architecture for different use cases while simulating the motor control.

6.4.1 Simulation of New Project

1. Make *MBCD_Transient_Simulation_Architecture* as current directory in the Matlab
2. Click on the *Simulink model MCU_Simulation_Architecture.slx*
3. Click on the New Project after pop-up window comes as shown below:

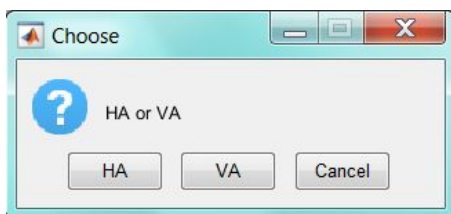


4. After New Project option is selected, another pop-up will come to select appropriate platform HA/VA

Page 21 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

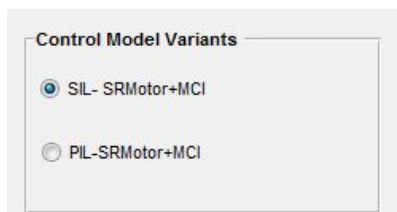
Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

as shown below:



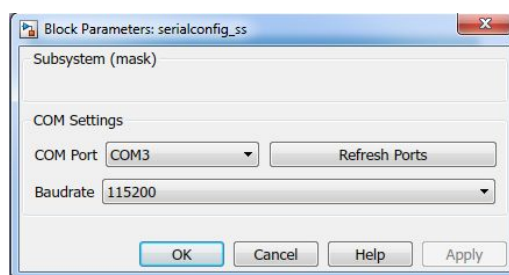
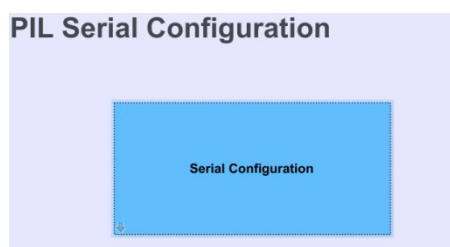
- After these initial selections, Simulink model will be opened along with model configuration UI. Model will be initialized depending upon the options selected in previous two steps.
- Select the Variant for the Control Simulation either SIL (Motor Control Source code is compiled as S-Function in the simulation) or PIL (Motor Control Source code is deployed on evaluation board)

Note: Model is initialized to the SIL as the Control Variant.



- In case, PIL variant is selected, following subsystem can be used to select COMPORT and to set Baud rate for the Processor:

PIL Serial Configuration



- Since, Motor Control Source code is simulated with SR+MCI, following options are provided to skip the initial delay in the simulation of the speed profile:

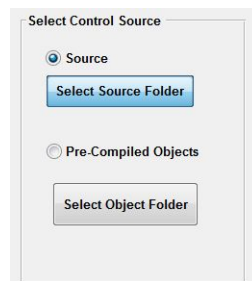
Page 22 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



Note: Model is initialized with the above options as checked.

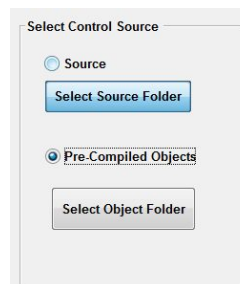
9. Select the Control algorithm source code by selecting appropriate source folder from particular project as shown below:



10. Alternately, pre-compiled objects can be selected by selecting the appropriate object folder as shown Below:

NOTE

By Default Object Files are stored at the folder **mcu_temp_obj**.



11. Select the frequency (Step-Size)of the simulation as shown below:

Page 23 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Sampling Frequency

kHz

12. For the Plants, by selecting the corresponding plant parameter file, parameters will be loaded in the Matlab base workspace as shown. In the below selection, BPM Plant Model will be parameterized for DDBPM Janus 5.0cuft.

Amotech.xml
DDBPM_Janus_4.5cuft.xml
DDBPM_Janus_5.0cuft.xml

13. CIM or BPM Motor can be selected from following controls:

Select Motor

☐ CIM ☒ BPM

14. Model fidelity for the Motor Plant is selected as shown below. Basic Model will invoke the basic formulation of BPM or CIM. Advanced Model will invoke the MBCD plant model FMUs in the architecture.

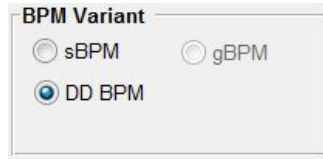
Model Fidelity

☒ Basic Model
☐ Advanced Model

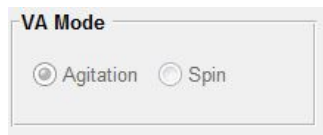
15. For BPM Motors, type of the BPM variant can be selected as shown below. Depending upon the selection, appropriate BPM parameter files will be available for selections. Note that by default gBPM will be invoked if VA platform is selected.

Page 24 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2011 7	Creation Date

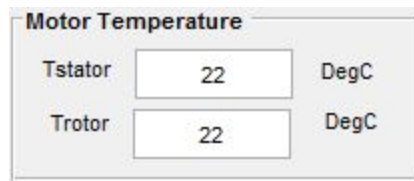
Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



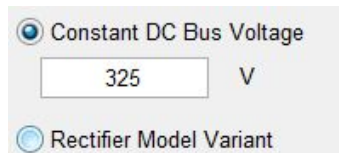
16. If VA platform is selected, gBPM motor model can have Agitation and spin mode variants that can be invoked from the controls shown below:



17. Stator and Rotor Temperature for both CIM and BPM motors can be set from the following controls. Architecture is initialized to the 22 DegC for both stator and rotor temperatures.



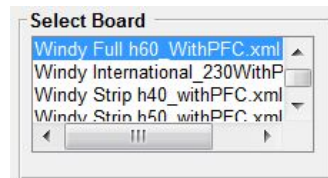
18. Input Circuit variant is selected from the controls shown in the figure below. Model is initialized to constant DC bus voltage of 325V.



19. **Note:** Board selection can be done independent of Input circuit variant so as to parameterize and simulate the ADC model for which parameters are board specific.

Page 25 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

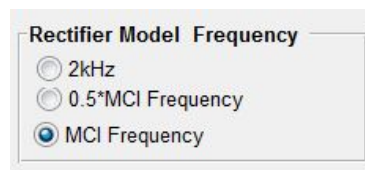


20. Rectifier model variant is selected with the help of control as shown in the figure below.

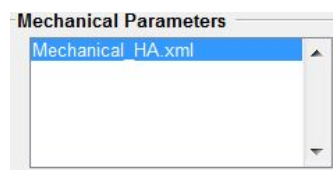


21. Note: With PFC variant is invoked only with 230V boards.

22. Rectifier variants can be simulated at different frequencies than the control simulation frequency with the help of controls shown in the following figure:



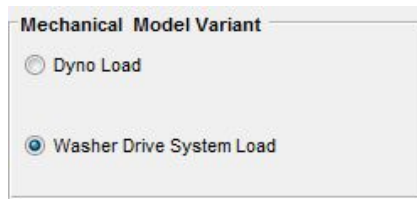
23. List of Mechanical parameter file selected will be as shown in the following figure:



Page 26 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

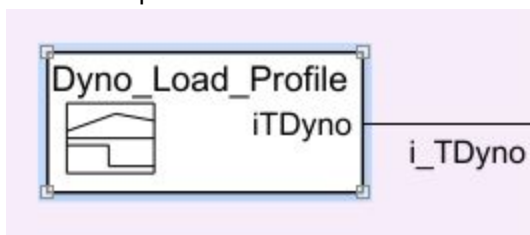
24. Mechanical Model Variant Can be changed with the help of following control:



Dyno Load option simulates Mechanical Model as simple load torque acting on the motor shaft.

NOTE (If Dyno Load option is invoked in the Simulation):

- Dyno Load Torque Option for the HA Basic Model variant is disabled as this functionality is modeled in the Advanced Motor model Variants.
- Dyno Load Torque Option for the VA Advanced Model variant is disabled.
- Dyno Load Torque Setting is initialized to zero.
- Dyno Load Torque Profile can be set with the help of following Signal generator block:



25. Drum Inertia can be changed by enabling the drum properties as shown in the following figure:

Page 27 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Drum Properties

☐ Enable
☒ Disable

Weight of Drum(M_D)
Kg

Weight of Inertial Mass(M)
Kg

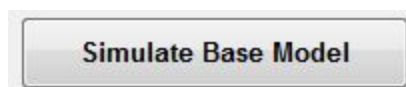
Radius of Drum(R_Drum)
m

Thickness of Inertial Mass (R_J)
m

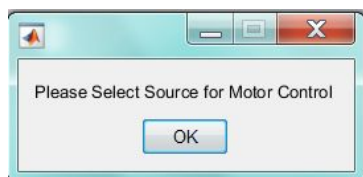
NOTE:

- Above option will be enabled only if Washer Drive System is selected as mechanical model variant.
- Basic Models for CIM and BPM have all of the parameters as plant parameters associated with the calculation of J_Drum. However, MBCD plant models of the BPM and CIM have only J_Drum as plant parameter hence, with the help of model configuration UI, J_Drum can be controlled with same granularity as with the Basic Models of BPM and CIM.

26. Once architecture is configured, motor control simulation can be started with the help of following control:



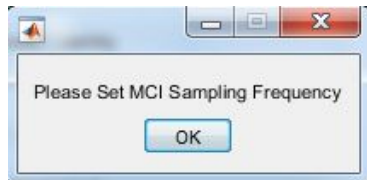
27. If the Motor Control Source Code folder is not selected, following message will appear if above control is clicked.



Page 28 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

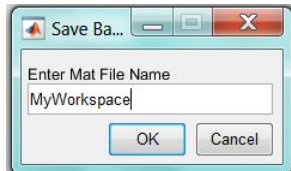
28. If the simulation step-size (or MCI Simulation Frequency) is not set, following message will appear:



29. Matlab base workspace can be saved as .MAT file for the future usage with the help of control as shown below:



30. Once the above control is clicked, following pop up will come to save base workspace:



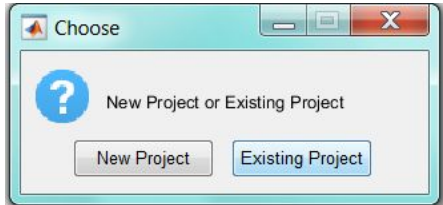
31. Note: Matlab Base workspace will be stored as .MAT file in the folder- Saved Workspace

6.4.2 Simulation of Existing Project

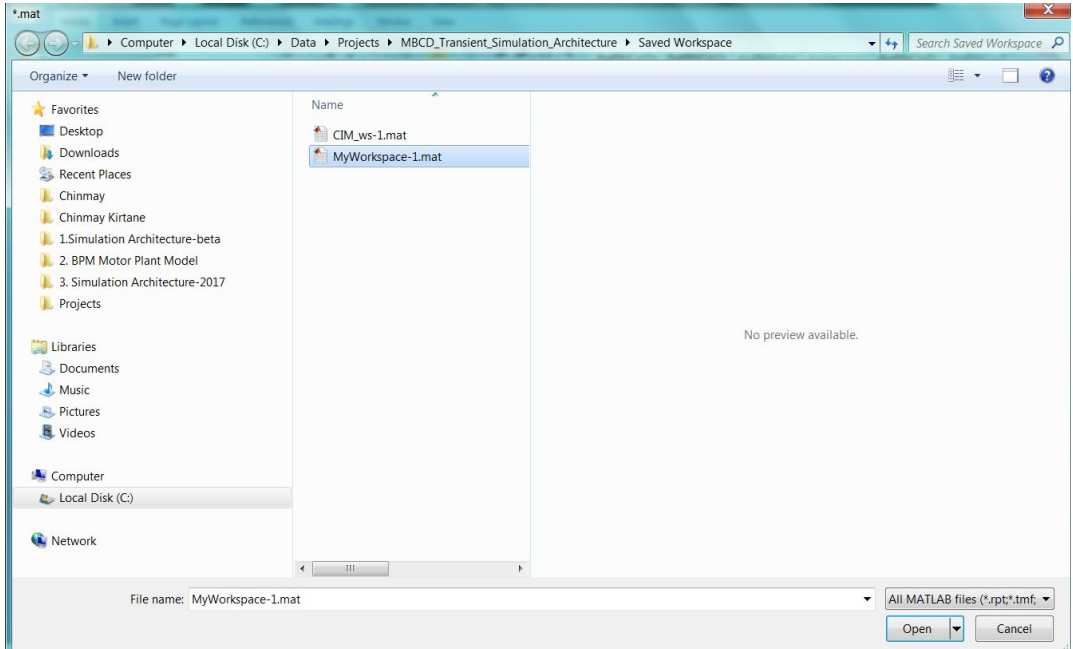
- Simulation architecture can also be simulated by loading previously stored workspace data.
- In order simulate in this mode, click on *Existing Project* as shown in the following table while opening the model-

Page 29 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



3. Once Existing Project is clicked, choose appropriate .MAT file from the Saved Workspace location as shown in the following figure:

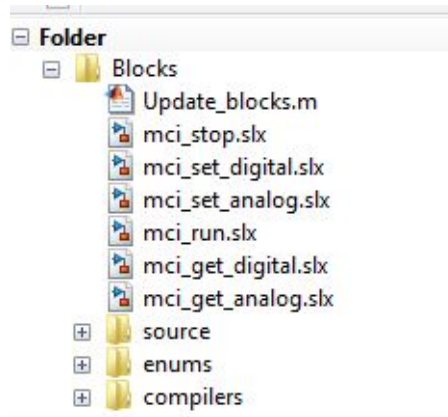


Page 30 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

6.5 MciAPI

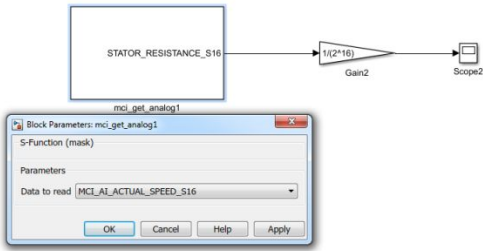
MciAPI Folder contains following modules as shown below:



- This Folder contains Models having S-Functions for the following methods:
 - MCI Run
 - MCI Stop
 - MCI Set Digital
 - MCI Set Analog
 - MCI Get Digital
 - MCI Get Analog.
- Above mentioned S-Functions can be used in the Architecture to send or retrieve the data from motor control source code.
- For e.g. mci_get_analog :

Page 31 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

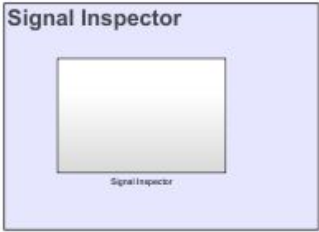
Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



4. Required Parameter can be set from the drop down menu available from the mask.

6.6 Signal Inspector

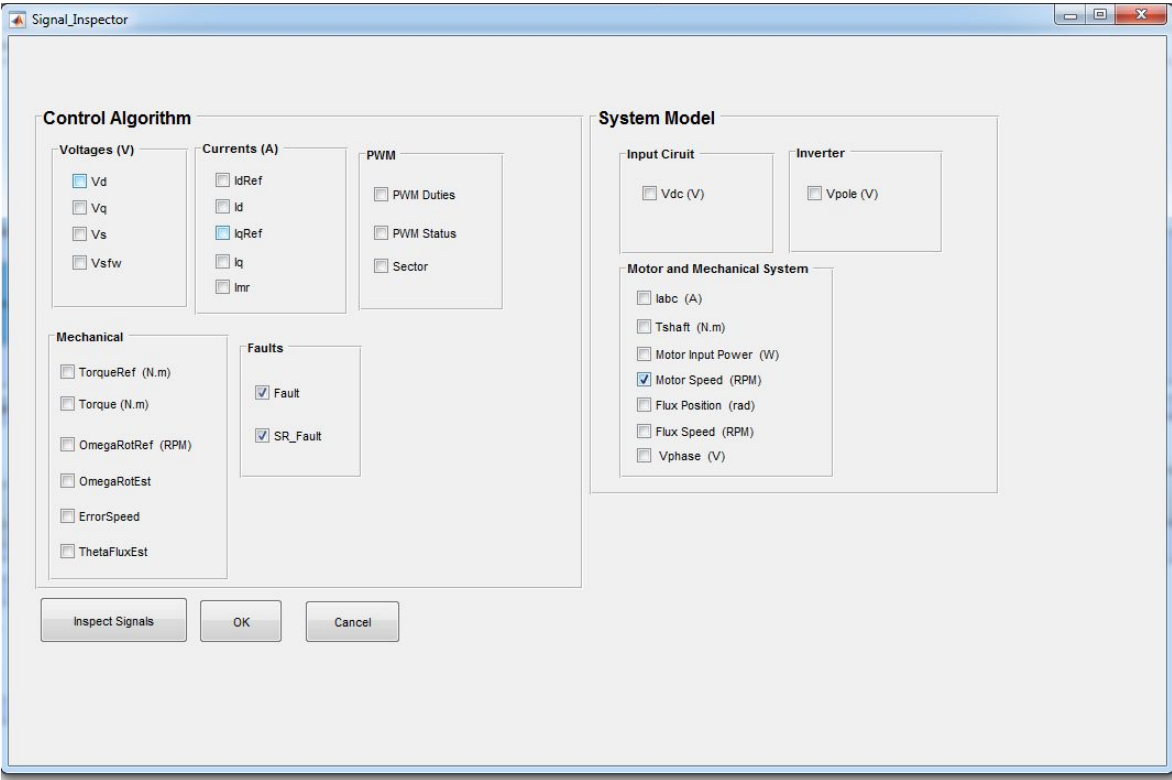
1. Signal Inspector sub-system can be used to stream the signal to the Simulink Data Inspector:



2. Signal Inspector GUI will be opened by clicking on Signal Inspector Sub-system:

Page 32 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

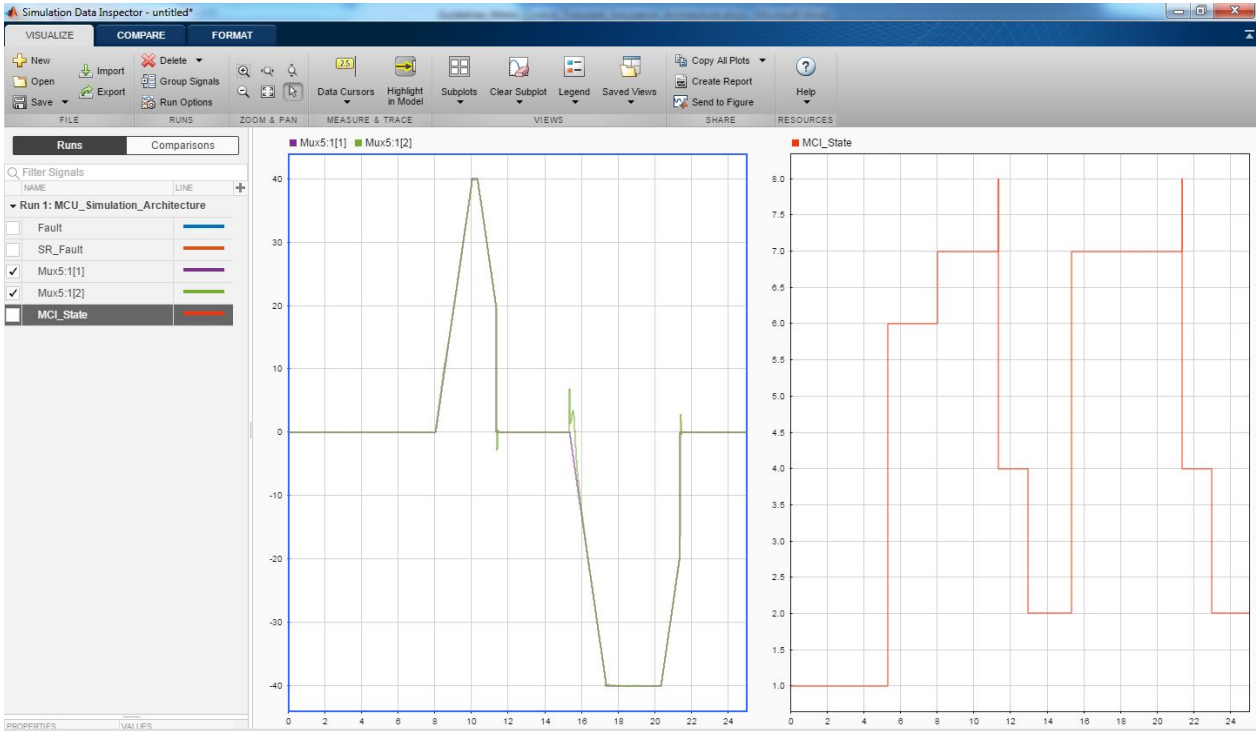
Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



3. Signals required to be streamed to the Simulink Data Inspector can be checked as shown in the figure above.
4. After the simulation, Simulink data inspector can be opened by clicking on Inspect Signals pushbutton:

Page 33 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



Page 34 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

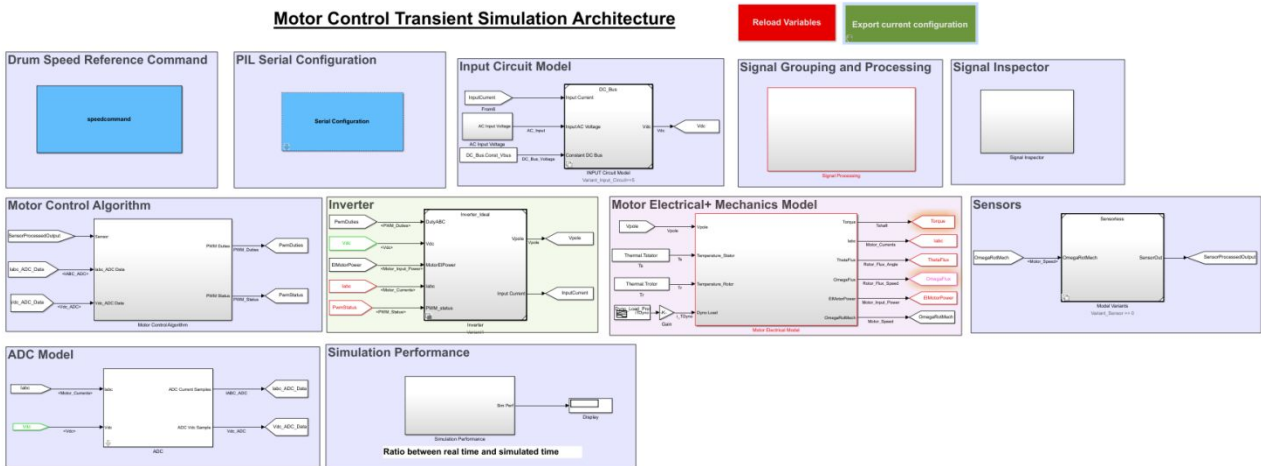
6.7 Exporting as Separate Model

Latest releases of the Simulation Architecture have an extra button to export the current configuration as an independent model.

One of the main reasons for this feature is to expose the architecture to external harness, making it easier to use the results provided by a particular configuration without having to make changes to the actual MCU_Simulation_Architecture.slx model.

After configuring the Simulation Architecture according to item 6.4 you can export such configuration as an atomic model by following these steps:

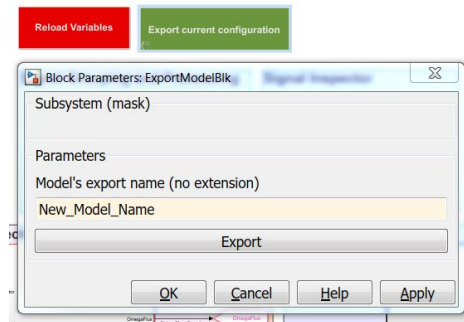
- Double click on the green button called 'Export current configuration'.



- Type in a name to the new model in the 'Model's export name (no extension)' box. Just a name, without model's file extension.

Page 35 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

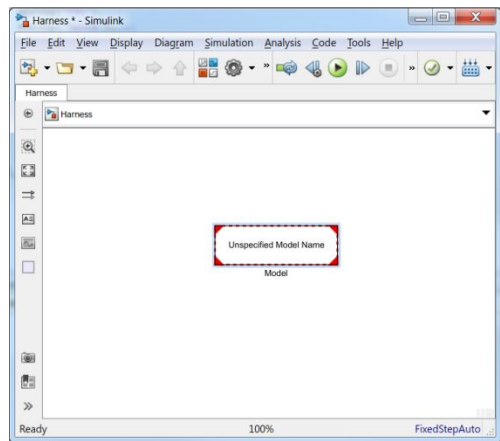


- Click 'Export' button.

The new model will be created under the folder 'Exported Models' located at your Simulation Architecture root folder.

6.7.1 Using Exported Model as Reference

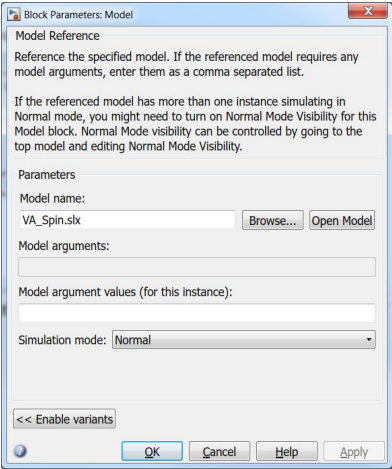
To use an exported model from a Simulation Architecture configuration, add a *Model block* into your new model.



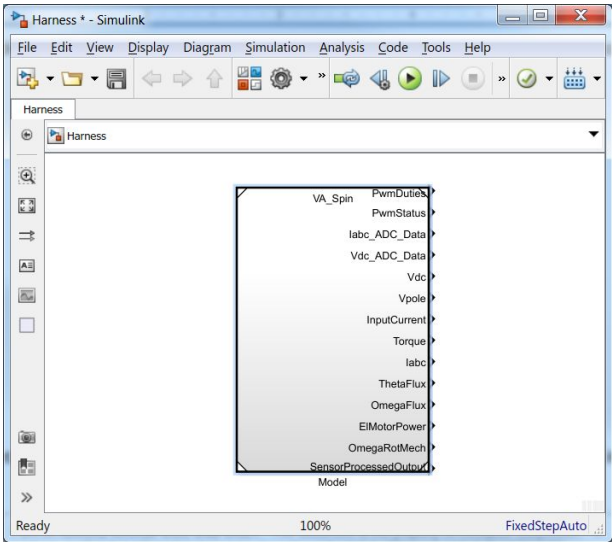
Double click the *Model block*, click on the 'Browse' button and select the exported model inside 'Exported Models' folder.

Page 36 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			



Once you click the ‘OK’ button, the exported model will be shown as a reference in your new model, as bellow:



Simulation values can be retrieved from the named output ports, for model’s variables, and using MCI API blocks, for MCI variables, as shown in item 6.5.

6.7.2 Reloading MCU_Simulation_Architecture variables

Page 37 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date

Document Number W1234	GUIDELINE	Document Type:	
	MOTOR CONTROL TRANSIENT SIMULATION ARCHITECTURE	Description (English)	
Revision A		Description (Local)	
Whirlpool Corporation, Benton Harbor, MI, USA			

Each exported model has its own configuration stored in its own *Model Workspace*.

Every time one of these models is run, it loads its own configuration into the *Base Workspace*, which is shared with MCU_Simulation_Architecture model.

If you try to run the MCU_Simulation_Architecture, that is already opened, after one of the exported models, there is a chance of loading values from the exported model that has just run.

To avoid that, when this situation happens, an error message is thrown and the simulation of the MCU_Simulation_Architecture model is halted.

To continue simulating the simulation architecture normally you must double click the red button called 'Reload Variables'.

Page 38 of 38	FOR INTERNAL WHIRLPOOL USE ONLY. This document is the property of Whirlpool Corporation. It contains confidential information belonging to and/or entrusted to Whirlpool and shall not be held, reproduced, disclosed outside of Whirlpool or used except in a manner and for a purpose expressly permitted by Whirlpool in writing.	KIRTACA	Created By
		26-September-2017	Creation Date